

Deniz KARAKOYUN

karakoyun.deniz@metu.edu.tr — karakoyundenizz.github.io
github.com/karakoyundenizz — linkedin.com/in/deniz-karakoyun

PROFILE

Computer Engineering student with a strong foundation in **Low-Level Programming, Computer Organization, Assembly, and Operating Systems (OS)**, with a keen interest in **Algorithm Design**. Experienced in C/C++, Assembly (x86-64), and developing high-throughput data pipelines with complex process synchronization. Passionate about bridging hardware and software to engineer reliable, architecture-aware systems.

EDUCATION

Middle East Technical University (METU)
B.Sc. in Computer Engineering

Sep 2021 – Expected 2027
CGPA: 3.61 / 4.00

Süleyman Demirel Anatolian High School
Graduated with 94.41 / 100

2017 – 2021

TECHNICAL SKILLS

Languages: C++, C, C#, Assembly (x86-64), Java, Python, SQL, Haskell, Verilog
Domains: Algorithms & Data Structures, Low-level Programming, Memory Management, OOP, Operating Systems, Computer Organization
Tools: Git, GDB, ROS 2, Kafka, ZeroMQ, MySQL, PostgreSQL, Docker, MATLAB

EXPERIENCE

Software Engineering Intern, KUARTIS

2026 – Present

- Contributing to a C++17 ROS 2 (Humble) lifecycle node that monitors GNSS signal health for an autonomous platform running on NVIDIA Jetson.
- Test-drove a signal-loss watchdog: wrote the failing integration test exposing a case where a silent GPS feed looked healthy, then implemented the freshness check and fix-quality state machine that close it.
- Fixed a data race between a DDS subscription callback and the test thread in a multi-threaded test harness.

Candidate Engineer, ASELSAN

Jul 2026 – Present

- Architecting data generation pipelines processing millions of data points for robust system evaluation and offline experimentation.
- Implementing advanced multi-threading and synchronization mechanisms to manage high-volume data streams without race conditions.

Undergraduate Researcher, ROMER

Jan 2026 – Present

Project: Honey Bee Orientation Detection

- Developing deep learning pipelines using CNN and ResNet architectures for precision computer vision tasks.
- Managing the full model lifecycle, from custom dataset generation to hyperparameter tuning for optimization.

Intern, BİTES

Aug 2025 – Sep 2025

- Developed modular C# desktop applications using WPF and the MVVM architectural pattern.
- Implemented complex data binding and validation logic to ensure high UI responsiveness.
- Conducted testing procedures to isolate service layers and ensure software stability.

Intern, ASELSAN

Jul 2025 – Aug 2025

- Developed C++ test scenarios using Google Test to verify the reliability of embedded systems.
- Automated GUI workflows for critical internal desktop tools using Python scripts (Pywinauto).

Intern, EXA4MIND (EU Research Project)

Aug 2024 – Sep 2024

- Architected low-latency streaming pipelines utilizing Kafka, Flink, and ZeroMQ for scientific data.
- Optimized fault-tolerance mechanisms to ensure data integrity during large-scale simulations.

PRODUCTS

Guild — Student Community App

[getguild.app](#) · App Store & Google Play

- Co-founded with Emirhan Güler (METU); sole developer of the product: verified university-email accounts, a forum

with anonymous posting and live polls, campus events with seat reservations and door check-in, and a nationwide opportunities board.

- Live in production at METU, Hacettepe, and Ankara University on iOS, Android, and the web.

Phera Labs — Digital Art Platform

pheralabs.com

- Sole developer of an art-tech startup's platform: a university-gated digital art gallery whose best works are curated into an international NFT collection.
- Built the full commerce stack: limited-edition print sales, QR-ticketed events with door check-in, referral rewards, and the admin back-office.

PROJECTS

Concurrent Data Pipeline Orchestrator

C/C++, Unix API

- Developed a robust process controller in C/C++ to orchestrate multi-stage CSV data processing pipelines utilizing Unix system calls (`fork`, `exec`, `pipe`, `dup2`).
- Engineered an n-ary tree process architecture, merging streams from recursive sub-merger nodes.
- Implemented strict inter-process communication (IPC) and synchronization using bidirectional Unix domain stream sockets (`socketpair`) to prevent deadlocks during concurrent operations.

Pipelined Processor Optimization

Assembly, HCL, Y86-64

- Extended a Y86-64 pipelined processor simulator by modifying Hardware Control Logic (HCL).
- Reduced Cycles Per Element (CPE) significantly by implementing loop unrolling, instruction reordering, and custom function design to optimize assembly benchmarks.

Bomb & Attack Labs

x86-64, GDB

- Reverse-engineered stripped x86-64 binaries stage by stage to defuse a multi-stage binary bomb; built code-injection and return-oriented-programming exploits in a sandboxed course lab.

Embedded Systems Control Simulator

Assembly (PIC18), MPLAB X

- Developed a concurrent system controller on a PIC18F8722 architecture, utilizing a round-robin approach to manage three independent state machines.
- Implemented precise I/O operations and hardware state management for multiple LED arrays (PORTB, PORTC, PORTD) without relying on hardware timers or interrupts within strict timing constraints.

Flight Finder System

C++, Graph Theory

- Developed a high-performance search engine using a Directional Multigraph and Hash Table.
- Implemented Dijkstra's algorithm with heuristics for cost-time blending and airline filtering.

REFERENCES

Available upon request.